

Exercises- Propositional Logic

Exercise 1

Check the following properties for $\downarrow$ ('nor'), $\uparrow$ ('nand') and $\oplus$ ('xor') connectives using the truth table method.

1. associativity of ' $\uparrow$ ' connective: $p \uparrow (q \uparrow r) \equiv (p \uparrow q) \uparrow r$;
2. associativity of ' $\downarrow$ ' connective: $p \downarrow (q \downarrow r) \equiv (p \downarrow q) \downarrow r$;
3. associativity of ' $\oplus$ ' connective: $p \oplus (q \oplus r) \equiv (p \oplus q) \oplus r$;
4. distribution of ' $\uparrow$ ' connective over ' $\downarrow$ ' connective:
$$p \uparrow (q \downarrow r) \equiv (p \uparrow q) \downarrow (p \uparrow r);$$
5. distribution of ' $\downarrow$ ' connective over ' $\uparrow$ ' connective:
$$p \downarrow (q \uparrow r) \equiv (p \downarrow q) \uparrow (p \downarrow r);$$
6. De Morgan's laws for ' $\downarrow$ ' and ' $\uparrow$ ':
$$\neg(p \downarrow q) \equiv \neg p \uparrow \neg q \quad \text{and} \quad \neg(p \uparrow q) \equiv \neg p \downarrow \neg q;$$
7. $p \uparrow (q \vee r) \equiv (p \uparrow q) \wedge (p \uparrow r)$ and $p \downarrow (q \wedge r) \equiv (p \downarrow q) \vee (p \downarrow r)$.
8. $p \downarrow (q \uparrow p) \equiv F$ and $p \uparrow (q \downarrow p) \equiv T$;

Exercise 2

Using the truth table method decide what kind of formula (consistent, inconsistent, tautology, contingent) is $U_j, j \in \{1, 2, \dots, 8\}$. Write all the models and anti-models of $U_j, j \in \{1, 2, \dots, 8\}$.

1. $U_1 = q \vee \neg p \vee r \rightarrow \neg(p \vee r)$;
2. $U_2 = \neg p \vee \neg(q \wedge r) \rightarrow q \wedge \neg p$;
3. $U_3 = \neg p \wedge (\neg q \vee r) \rightarrow q \wedge \neg p \vee r$;
4. $U_4 = \neg(\neg p \wedge q) \vee r \rightarrow p \wedge (\neg q \vee r)$;
5. $U_5 = \neg p \vee q \wedge r \rightarrow q \wedge \neg p$;
6. $U_6 = \neg p \vee (\neg q \wedge \neg r) \rightarrow q \wedge \neg p \wedge r$;
7. $U_7 = p \rightarrow (p \wedge r) \vee q$;
8. $U_8 = (p \vee q) \wedge \neg r \rightarrow p \wedge q \wedge r$.

Exercise 3

Using the truth table method, check whether the following logical consequences hold:

1. $p \rightarrow q \models (p \rightarrow r) \rightarrow (p \rightarrow q \wedge r)$;
2. $p \rightarrow q \models (q \rightarrow r) \rightarrow (p \rightarrow r)$;
3. $p \rightarrow (q \rightarrow r) \models (p \rightarrow q) \rightarrow (p \rightarrow r)$;
4. $p \rightarrow r \models (q \rightarrow r) \rightarrow ((p \vee q) \rightarrow r)$;
5. $p \rightarrow q \models (\neg p \rightarrow q) \rightarrow q$;
6. $p \rightarrow q \models (q \rightarrow r) \rightarrow (p \rightarrow q \wedge r)$;
7. $p \rightarrow q \models (q \rightarrow r) \rightarrow (p \rightarrow q \vee r)$;
8. $r \rightarrow (q \rightarrow p) \models (r \rightarrow q) \rightarrow (r \rightarrow p)$.

Exercise 4.

Prove that the following formulas are tautologies using the truth table method.

1. the left-distribution of ' $\rightarrow$ ' over ' $\wedge$ ': $(p \rightarrow (q \wedge r)) \rightarrow ((p \rightarrow q) \wedge (p \rightarrow r))$;
2. the permutation of the premises law: $(p \rightarrow (q \rightarrow r)) \rightarrow (q \rightarrow (p \rightarrow r))$;
3. the reunion of the premises law: $(p \rightarrow (q \rightarrow r)) \rightarrow (p \wedge q \rightarrow r)$;
4. the separation of the premises law: $(p \wedge q \rightarrow r) \rightarrow (p \rightarrow (q \rightarrow r))$;
5. the left-distribution of ' $\vee$ ' over ' $\leftrightarrow$ ': $(p \vee (q \leftrightarrow r)) \rightarrow ((p \vee q) \leftrightarrow (p \vee r))$;
6. the left-distribution of ' $\vee$ ' over ' $\rightarrow$ ': $p \vee (q \rightarrow r) \rightarrow ((p \vee q) \rightarrow (p \vee r))$;
7. the left-distribution of ' $\rightarrow$ ' over ' $\leftrightarrow$ ': $(p \rightarrow (q \leftrightarrow r)) \rightarrow ((p \rightarrow q) \leftrightarrow (p \rightarrow r))$;
8. the left-distribution of ' $\rightarrow$ ' over ' $\vee$ ': $(p \rightarrow (q \vee r)) \rightarrow ((p \rightarrow q) \vee (p \rightarrow r))$.

Exercise 5

Transform the formulas $U_j, j \in \{1, 2, \dots, 8\}$ into their equivalent conjunctive and disjunctive normal forms.

Using one of these forms prove that $U_j, j \in \{1, 2, \dots, 8\}$ are valid formulas in propositional logic.

1. $U_1 = (p \rightarrow \neg q) \wedge (q \vee r) \rightarrow (p \rightarrow r)$;
2. $U_2 = (p \rightarrow q) \wedge (p \wedge q \rightarrow r) \rightarrow (p \rightarrow r)$;
3. $U_3 = (p \wedge q \rightarrow r) \rightarrow (p \rightarrow (q \rightarrow r))$;
4. $U_4 = (p \rightarrow (q \vee r)) \rightarrow ((p \rightarrow q) \vee (p \rightarrow r))$;
5. $U_5 = (p \rightarrow q) \wedge (p \wedge q \rightarrow r) \rightarrow (p \rightarrow r)$;
6. $U_6 = (p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$;
7. $U_7 = (p \rightarrow (q \wedge r)) \rightarrow ((p \rightarrow q) \wedge (p \rightarrow r))$;
8. $U_8 = p \vee (q \rightarrow r) \rightarrow ((p \vee q) \rightarrow (p \vee r))$.

Exercise 6

Using the appropriate normal form write all the models of the following formulas:

1. $U_1 = (p \vee q \rightarrow r) \rightarrow (p \rightarrow r) \wedge q$;
2. $U_2 = \neg(\neg p \vee q) \vee r \rightarrow \neg p \wedge \neg(q \wedge r)$;
3. $U_3 = (p \wedge q \rightarrow r) \rightarrow (p \rightarrow r) \wedge q$;
4. $U_4 = (p \vee q) \wedge \neg r \rightarrow p \wedge q \wedge r$;
5. $U_5 = p \vee \neg(q \wedge \neg r) \rightarrow p \wedge q \wedge \neg r$;
6. $U_6 = (p \vee q \rightarrow r) \rightarrow (q \rightarrow r) \wedge p$;
7. $U_7 = (q \vee r \rightarrow p) \rightarrow (p \rightarrow r) \wedge q$;
8. $U_8 = (q \wedge r \rightarrow p) \rightarrow (p \rightarrow r) \wedge q$.

Exercise 7

Using the appropriate normal form, prove that the following formulas are inconsistent:

1. $U_1 = (p \rightarrow (q \rightarrow r)) \wedge \neg((p \rightarrow q) \rightarrow (p \rightarrow r))$;
2. $U_2 = (\neg p \vee q) \wedge \neg(\neg q \rightarrow \neg p)$;
3. $U_3 = (p \rightarrow q) \wedge (p \wedge q \rightarrow r) \wedge (p \wedge \neg r)$;
4. $U_4 = (p \rightarrow (q \vee r)) \wedge (\neg(p \rightarrow q) \wedge \neg(p \rightarrow r))$;
5. $U_5 = p \wedge (q \rightarrow r) \wedge ((p \wedge q) \wedge \neg(p \wedge r))$;
6. $U_6 = (p \rightarrow (q \rightarrow r)) \wedge (p \wedge q \wedge \neg r)$;
7. $U_7 = (p \rightarrow (q \rightarrow r)) \wedge \neg(q \rightarrow (p \rightarrow r))$;
8. $U_8 = (p \wedge q \rightarrow r) \wedge \neg(p \rightarrow (q \rightarrow r))$.

Exercise 8

Write all the anti-models of the following formulas using CNF.

1. $U_1 = (q \wedge r \rightarrow p) \rightarrow (p \rightarrow r) \wedge q$;
2. $U_2 = (q \vee r \rightarrow p) \rightarrow (p \rightarrow r) \wedge q$;
3. $U_3 = (p \vee q \rightarrow r) \rightarrow (q \rightarrow r) \wedge p$;
4. $U_4 = p \vee \neg(q \wedge \neg r) \rightarrow p \wedge q \wedge \neg r$;
5. $U_5 = p \wedge \neg(q \wedge \neg r) \rightarrow p \vee q \wedge \neg r$;
6. $U_6 = (p \wedge q \rightarrow r) \rightarrow (p \rightarrow r) \wedge q$;
7. $U_7 = \neg(\neg p \vee q) \vee r \rightarrow \neg p \wedge \neg(q \wedge r)$;
8. $U_8 = (p \vee q \rightarrow r) \rightarrow (p \rightarrow r) \wedge q$.

Exercise 9

Using the definition of deduction, prove the following deductions:

1. $p \rightarrow q, r \rightarrow t, p \vee r, \neg q \vdash t$;
2. $p \rightarrow r, p \vee r \rightarrow q, r \vdash q$;
3. $q \rightarrow p, t \rightarrow r, q \vee t, \neg p \vdash r$;
4. $p \vee (q \rightarrow r), p \vee q, \neg p \vdash r$;
5. $\neg p \vee \neg q \vee r, q, p \vdash r$;
6. $p \rightarrow \neg q \vee r, p \wedge q, p \vdash r$;
7. $r \vee (q \rightarrow p), r \vee q, \neg r \vdash p$;
8. $p \rightarrow q, q \rightarrow r, r \rightarrow t, p \vdash t$.

Exercise 10

Prove the following theorems using the theorem of deduction and its reverse.

1. $\vdash p \vee (q \rightarrow r) \rightarrow ((p \vee q) \rightarrow (p \vee r))$;
2. $\vdash (p \rightarrow (\neg r \rightarrow q)) \rightarrow (r \vee \neg p \vee q)$;
3. $\vdash (p \rightarrow (q \rightarrow r)) \rightarrow (p \wedge q \rightarrow r)$;
4. $\vdash (p \wedge q \rightarrow r) \rightarrow (p \rightarrow (q \rightarrow r))$;
5. $\vdash (p \rightarrow (q \rightarrow r)) \rightarrow (q \rightarrow (p \rightarrow r))$;
6. $\vdash (p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$;
7. $\vdash (p \rightarrow q) \wedge (p \wedge q \rightarrow r) \rightarrow (p \rightarrow r)$;
8. $\vdash (p \rightarrow q) \rightarrow ((p \rightarrow r) \rightarrow (p \rightarrow q \wedge r))$.

Exercise 11

Using the theorem of deduction and its reverse prove that:

1. $\vdash (p \rightarrow (q \vee r)) \rightarrow ((p \rightarrow q) \vee (p \rightarrow r))$;
2. $\vdash (p \rightarrow q) \rightarrow ((\neg r \vee p) \rightarrow (r \rightarrow q))$;
3. $\vdash p \vee (q \rightarrow r) \rightarrow ((p \vee q) \rightarrow (p \vee r))$;
4. $\vdash (p \rightarrow r) \rightarrow ((q \rightarrow r) \rightarrow (p \vee q \rightarrow r))$;
5. $\vdash (p \rightarrow q) \rightarrow ((r \rightarrow t) \rightarrow (p \wedge r \rightarrow q \wedge t))$;
6. $\vdash (p \rightarrow r) \rightarrow ((p \wedge r \rightarrow q) \rightarrow (p \rightarrow q))$;
7. $\vdash (\neg q \vee p) \rightarrow ((s \rightarrow q) \rightarrow (s \rightarrow p))$;
8. $\vdash (p \rightarrow (q \rightarrow r)) \rightarrow (p \rightarrow (\neg r \rightarrow \neg q))$.

Exercise 12

H1: It is not sunny this afternoon and it is colder than yesterday.

H2: We will go swimming only if it is sunny.

H3: If we do not go swimming, then we will take a canoe trip.

H4: If we take a canoe trip, then we will be home by sunset.

C: We will be home by sunset.

Is C deducible from the set of hypotheses {H1,H2,H3,H4}?

If yes, build its deduction.